

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **PLC programming**, **robotics integration**, and **vision system configuration** for automated production lines, reducing manual error rates by **18%**.
- Managed the full project lifecycle for automation upgrades, including **hardware selection** and **architecture design**, ensuring delivery within **schedule and budget** constraints.
- Developed and maintained **SCADA system support** for a live production environment, actively participating in a team to perform **fault finding** and **troubleshooting** of automation issues.
- Authored comprehensive **GAMP documentation** (URS, FDS, FAT, SAT) in collaboration with project teams, ensuring compliance with pharmaceutical manufacturing standards.
- Provided **on-call automation support** for critical production equipment, performing complex **fault finding** by interrogating program code to identify and resolve issues within **4-hour SLAs**.
- Coached junior control engineers on **PLC troubleshooting** and **process improvement** initiatives, contributing to enhanced machine reliability and productivity across the Biotoools department.

SKILLS

Automation & Control – PLC Programming, Robotics, Vision Systems, SCADA, Servo Drives, Motion Control, Industrial Network Protocols (EtherNet/IP, DeviceNet, Modbus, Profibus)

Programming & Scripting – Java, Spring Boot, Python, SQL

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

DevOps & CI/CD – Jenkins, Git, CI/CD Pipelines

Data & Analytics – Power BI, DAX, Power Query, Tableau, R, pandas, NumPy, scikit-learn

Manufacturing Systems – MES Integration, ERP Systems, Change Management, GAMP Documentation (URS, FDS, FAT, SAT)

Project Management – Capital Project Justification, Budget Management, Requisition Management, Contract Negotiation

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Automated Quality Control Vision System

- Developed a **Python-based vision system** for automated product inspection, integrating with existing **PLC controls** and **servo motors** to ensure consistent quality checks.

- Configured **Cognex vision system** parameters and developed custom logic to identify defects, improving detection accuracy by **22%** and reducing manual inspection time.
- Managed the installation and commissioning of the vision system, including data logging to **AWS S3** for historical analysis and reporting.

Production Line Performance Dashboard

- Designed and implemented an interactive **Power BI dashboard** to monitor key performance indicators for a high-speed production line, integrating data from **SCADA** and MES systems.
- Developed complex **DAX measures** to calculate real-time OEE (Overall Equipment Effectiveness), throughput, and downtime, providing actionable insights to operations teams.
- Automated data refresh processes using **Power Query**, ensuring stakeholders had access to up-to-date performance metrics for informed decision-making.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)